

Session 211: Speech and Sensor Signal Processing

TIME 9:30 AM – 11:00 AM IST VENUE SP 1.20 (TVS Lab)



211.1

RASE: A Recursive-Adaptive Spectral Enhancer for Multilingual Indian Language Speech Enhancement

Satya Prasad Gaddamedi, Debolina Pramanik, Puja Bharati, Sabyasachi Chandra, Shyamal Kumar Das Mandal, Nirmalya Ghosh, Satya Prasad Gaddamedi

211.2

Tracking Phonetic Transitions in Self-Supervised Speech Representations

Kumar Kaustubh, Aastha Kachhi, Ashish Issac, Ajay Kumar Sharma

211.3

Overcoming NAWEs Fixed-Window Constraints For Spoken Term Detection And Localization

Ramesh Gundluru, Sreekanth Sankala, Venkatesh Parvathala, Sri Rama Murty Kodukula

211.4

Semi-Supervised FFT-Spectral Representation Learning for Earthquake Detection in Distributed Fiber Optic Sensing Systems

Deepika Sasi, Sundaresan Sabapathy, Thomas Joseph

211.5

Direction of arrival estimation using subspace based iterative orthogonal projection

Praveen Babu Choppala, Mohammad Sajida Khan, Swarnim Rishal

211.6

Real-Time Correction of Named Entities in Speech Transcriptions: A Lightweight Approach

Kiranmayi Gandikota, Anunay Katare, Chandan Kumar Pandey, Mayank Kumar Tyagi, Subhadarshi Panda, Rahul Kumar, Mohamed Asfak R

211.7

Perceiver encoder for automatic speech recognition

Kaustubh Kulkarni, Venkateswarlu Gonuguntla

211.8

Phoneme-Preserving Mamba-Based Frontend for Noise-Robust Automatic Speech Recognition

Reshma K R, Anil Kumar Vuppala

211.9

Spoken Language Diarization using Source-Domain Aggregative Weakly-Supervised Language Labels

Spandan Dey, Hirak Mondal, Anmol Jain, Shirish Kumar, Premjeet Singh, Suyash Pandya

211.10

Exploiting Compressed Eigencomponent Transformation in DNN Framework for Passive SONAR Self-noise Cancellation

Pawan Kumar, Karan Nathwani

211.11

Impact of Localization Errors on Deep Learning Models for Wideband Spectrum Monitoring

Aman Kumar, Jay Vikrant, Sundar Vanka, Sumohana Channappayya

211.12

QMOS: Qwen-based MOS Prediction for TTS

Ravi Sastry Kolluru, Sri Charan Devarakonda, Radhe Shyam Salopanthula, Anil Kumar Vuppala

211.13

Validating Smartphone-Based 3D Head Models for Precision EEG Source Localization

Suranjita Ganguly, Malaika Mihir Chhaya, Ankita Jain, Aditya Koppula, Mohan Raghavan, Kousik Sarathy Sridharan

211.14

Audio-Based Urgency Detection with SHAP-Enhanced Feature Refinement

Ankit Dighe, Sanika Gadkarl

211.15

Leveraging Deep Bayesian Inference for Efficient Multiple Source Localization with Spherical Harmonics

Priyadarshini Dwivedi, Gyanajyoti Routray, Rajesh M. Hegde

Session 212: Secure and Terahertz Communication Systems

TIME 9:30 AM – 11:00 AM IST VENUE SP Building Side Foyer



- 212.1** **Variational Bayes Estimation for Affine-Precoded Superimposed Pilots in Partially Connected Dual-Wideband Terahertz MU-MIMO Systems**
Abhisha Garg, Suraj Srivastava, Aditya Jagannatham
- 212.2** **Improving the Cheater Identification Capability of Threshold Secret Sharing Schemes**
Tanmoy Mandal, Srinivasan Krishnaswamy, Ratnajit Bhattacharjee
- 212.3** **Finite-Key Analysis of Phase-Matching QKD Type Protocol over a Satellite Channel under Atmospheric Turbulence**
Pranav Surapuraju, Lokendra Chouhan
- 212.4** **End-to-End Post-Processing for CV-QKD with Heterodyne Detection: ADC Modelling, Amplitude Sifting, Tri-Mode NB-LDPC Decoding, and Selective Privacy Amplification**
Mohammad Sikander sheikh, Rohit Chaurasiya
- 212.5** **EM-based Model for RIS-aided Channel and its Achievable DoF**
Nobhendu Chowdhury, Praful Mankar
- 212.6** **Regularized Multi-Measurement Sparse Decomposition for Time-Domain Ambisonics Upscaling**
Gyanajyoti Routray, Priyadarshini Dwivedi, Rajesh M. Hegde
- 212.7** **Terahertz Topological Valley Photonic Crystals with Polymer-Filled Bearded Interface for Robust Waveguiding in 6G Applications**
Dibaskar Biswas, Tushar Shah, Sourodipto Das, Amit Baran Dey, Ashish Kumar Chowdhary, Debabrata Sikdar
- 212.8** **Topologically Protected Supercoupling in Cavity-Integrated Photonic Structure for Terahertz Communication**
Tushar Shah, Dibaskar Biswas, Lokesh Sharma, Sourodipto Das, Ashish Kumar Chowdhary, Debabrata Sikdar

Session 213: Signal and Image Processing

TIME 9:30 AM – 11:00 AM IST VENUE ECE 1.07



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213.1

Convolution Based Fast Lossless Compression Algorithm for Color Filter Arrays

Chiranjeev Bhaya, Manish Kumar, Chandra Mohan Velpula, Praveen Bangre Prabhakar Rao

213.2

Modelling Bandlimited Signals in Self-Dual Differential Equations

Abhijat Bharadwaj, Animesh Kumar, Shaileshram Sivakumar, Advay Bapat, Vikram Gadre, Shrikant Sharma, P. Radhakrishna, Ram Bilas Pachori

213.3

Hierarchical Convolution Spatial Propagation Networks for Depth Completion

Praveen Ramesh, Vikas Gowda, Kunal A Kathare, Basavaraja Shanthappa Vandrotti, Jooyoung Kim, Ankit Dhiman

213.4

Bilevel Learning of Task-Adapted Regularizers

Aryan Saha, Subhadip Mukherjee

213.5

Consecutive Support Matching Induced Parameter Tuning Accelerates Momentum Iterative Hard Thresholding

Samrat Mukhopadhyay, Debasmita Mukherjee

213.6

Physics-Guided Residual-Aware Consensus Equilibrium for SAR Image Despeckling

Satyakam Baraha, Abhijit Mishra, Ajit Kumar Sahoo

213.7

Consistency is Key to Image Denoising

Oindrila Haldar, Saptarshi Mandal, Chandra Sekhar Seelamantula

213.8

Accelerating Quaternion Low-Rank Approximation for Color Images via Complex Embedding

Akshadha A, Devika Rani J, Sooraj K. Ambat

213.9

Hard Thresholding Based LMS Algorithm for Adaptive Graph Signal Recovery

Rachita Mohanty, Rajib Lochan Das

213.10

Mixing Time Bounds for Asymmetric Random Walks on Hypercubes, Cycles, and Tori Using Coupling

Mrudula Mahindrakar, Hrushikesh Kant, Avhishek Chatterjee

213.11

A Small-World Network Based Framework for Texture Classification

Shubham Dwivedi, Keerthika Kadagala, Om Jee Pandey, Rajesh M. Hegde

213.12

Power Series Based Iterative Algorithms for Distributed Sparse Recovery

Shashank S, Ketan Atul Bapat, Mrityunjay Chakraborty